

2014 年真题参考答案

一、选择题

(1) B. (2) C. (3) D. (4) C. (5) D. (6) A. (7) B. (8) A.

二、填空题

(9) $\frac{3\pi}{8}$. (10) 1. (11) $-\frac{1}{2}(\mathrm{d}x + \mathrm{d}y)$. (12) $\frac{2}{\pi}x + y - \frac{\pi}{2} = 0$. (13) $\frac{11}{20}$. (14) $[-2, 2]$.

三、解答题

(15) $\frac{1}{2}$.

(16) 极大值 $y(1) = 1$, 极小值 $y(-1) = 0$.

(17) $-\frac{3}{4}$.

(18) $f(u) = \frac{1}{16}e^{2u} - \frac{1}{16}e^{-2u} - \frac{1}{4}u$.

(19) 证明略.

(20) 1.

(21) $\left(2\ln 2 - \frac{5}{4}\right)\pi$.

(22) (I) $(-1, 2, 3, 1)^T$.

(II) $\mathbf{B} = \begin{pmatrix} 2 - k_1 & 6 - k_2 & -1 - k_3 \\ -1 + 2k_1 & -3 + 2k_2 & 1 + 2k_3 \\ -1 + 3k_1 & -4 + 3k_2 & 1 + 3k_3 \\ k_1 & k_2 & k_3 \end{pmatrix}$, 其中 k_1, k_2, k_3 为任意常数.

(23) 证明略.